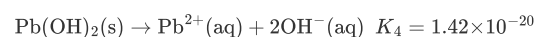
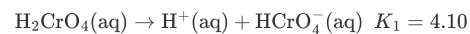


Question

Known are the following data:



Excess $\text{PbCrO}_4(\text{s})$ is added to pure water, and solution equilibrium is reached.

Please calculate the following questions in order:

1. The concentration of H_2CrO_4 in the solution in units of $\text{mol} \cdot \text{L}^{-1}$.
2. The solubility of $\text{PbCrO}_4(\text{s})$, expressed in terms of Cr element, in units of $\text{mol} \cdot \text{L}^{-1}$.
3. At this time, the relationship between the amount of substance of $\text{Pb}(\text{OH})_2(\text{s})$ and the volume of water satisfies a direct proportional function $n = aV$, where n is the amount of substance (in units of mol), and V is the volume of water (in units of L). Give the coefficient a (no need to give units).

The following options give possible answers to questions 1, 2, and 3 in order. The units have already been given in the questions. Choose the closest option.

A. 1.82×10^{-14} , 1.67×10^{-6} , 8.47×10^{-15}

B. 3.57×10^{-11} , 6.07×10^{-7} , 8.13×10^{-7}

C. 1.52×10^{-15} , 5.81×10^{-7} , 2.48×10^{-9}

D. 3.02×10^{-7} , 1.67×10^{-6} , 8.47×10^{-15}

E. 4.10×10^{-15} , 6.58×10^{-7} , 4.0×10^{-9}

F. 2.01×10^{-15} , 5.83×10^{-7} , 4.2×10^{-9}

Answer

Correct Answer: C

Detailed Explanation

The charge conservation equation:

$$[H^+] + 2[Pb^{2+}] = [OH^-] + [HCrO_4^-] + 2[CrO_4^{2-}] + 2[Cr_2O_7^{2-}]$$

CHECKPOINT

1 PTS

$$[H^+] + 2[Pb^{2+}] = [OH^-] + [HCrO_4^-] + 2[CrO_4^{2-}] + 2[Cr_2O_7^{2-}]$$

$$[H_2CrO_4] = (K_3K_w^2)/(K_1K_2K_4) = (2.80 \times 10^{-13} \times 10^{-14 \times 2}) / (4.10 \times 3.16 \times 10^{-7} \times 1.42 \times 10^{-20}) = 1.52 \times 10^{-15}$$

CHECKPOINT

3 PTS

$$[H_2CrO_4] = 1.52 \times 10^{-15}$$

Substituting the expressions of the equilibrium constants into the charge conservation equation, it is rewritten as:

$$[H^+] + (2K_4[H^+]^2)/K_w^2 = K_w/[H^+] + (K_1[H_2CrO_4])/[H^+] + (2K_1K_2[H_2CrO_4])/[H^+]^2 + (2K_1^2[H_2CrO_4]^2)/(K_5[H^+]^2)$$

Solving the above equation yields: $[H^+] = 6.39 \times 10^{-8} \text{ mol} \cdot L^{-1}$

CHECKPOINT

3 PTS

$$[H^+] = 6.39 \times 10^{-8} \text{ mol} \cdot L^{-1}$$

The solubility of $PbCrO_4$, measured by the total chromium concentration in the solution, is given by:

$$c = [H_2CrO_4] + [HCrO_4^-] + [CrO_4^{2-}] + 2[Cr_2O_7^{2-}] = [H_2CrO_4] + (K_1[H_2CrO_4])/[H^+] + (K_1K_2[H_2CrO_4])/[H^+]^2 + (2K_1^2[H_2CrO_4]^2)/(K_5[H^+]^2) = 5.81 \times 10^{-7} \text{ mol} \cdot L^{-1}$$

CHECKPOINT

3 PTS

$$c = 5.81 \times 10^{-7} \text{ mol} \cdot L^{-1}$$

The concentration of $Pb(OH)_2$ equals the concentration of dissolved Pb (i.e., solubility c) minus the concentration of Pb^{2+} , leading to the following equation:

$$[Pb(OH)_2] = c - [Pb^{2+}] = c - (K_4[H^+]^2)/K_w^2 = 2.48 \times 10^{-9} \text{ mol} \cdot L^{-1}$$

CHECKPOINT

1 PTS

$$[Pb(OH)_2] = c - [Pb^{2+}] = c - (K_4[H^+]^2)/K_w^2 = 2.48 \times 10^{-9} \text{ mol} \cdot L^{-1}$$

Thus, the functional relationship is:

$$a = 2.48 \times 10^{-9}$$

CHECKPOINT

1 PTS

$$a = 2.48 \times 10^{-9}$$